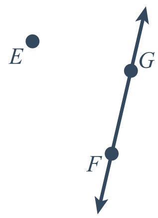
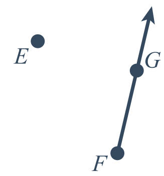
Review: Points, lines, and planes

1 Name the line, line segment, or ray in each diagram below using geometric notation.

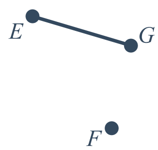
a



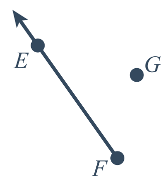
b



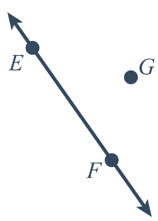
c



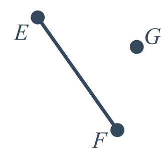
d



e

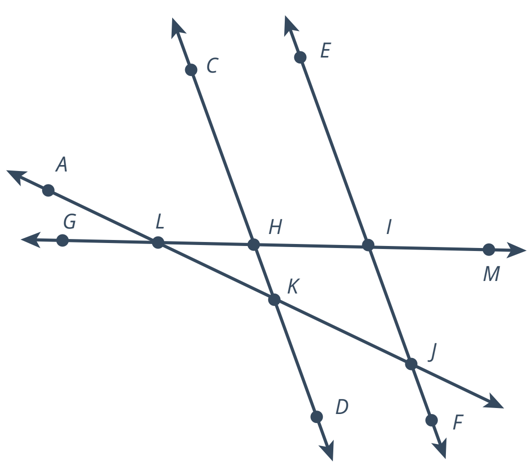


f

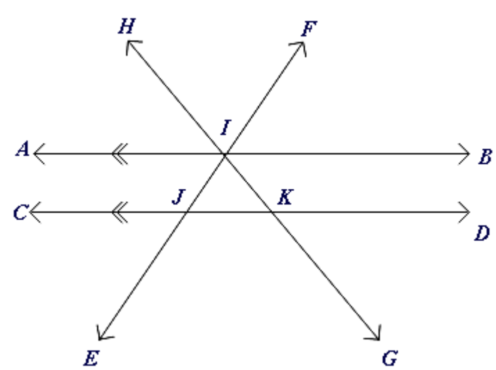


2 Name three different pairs of collinear points in each diagram below.

a

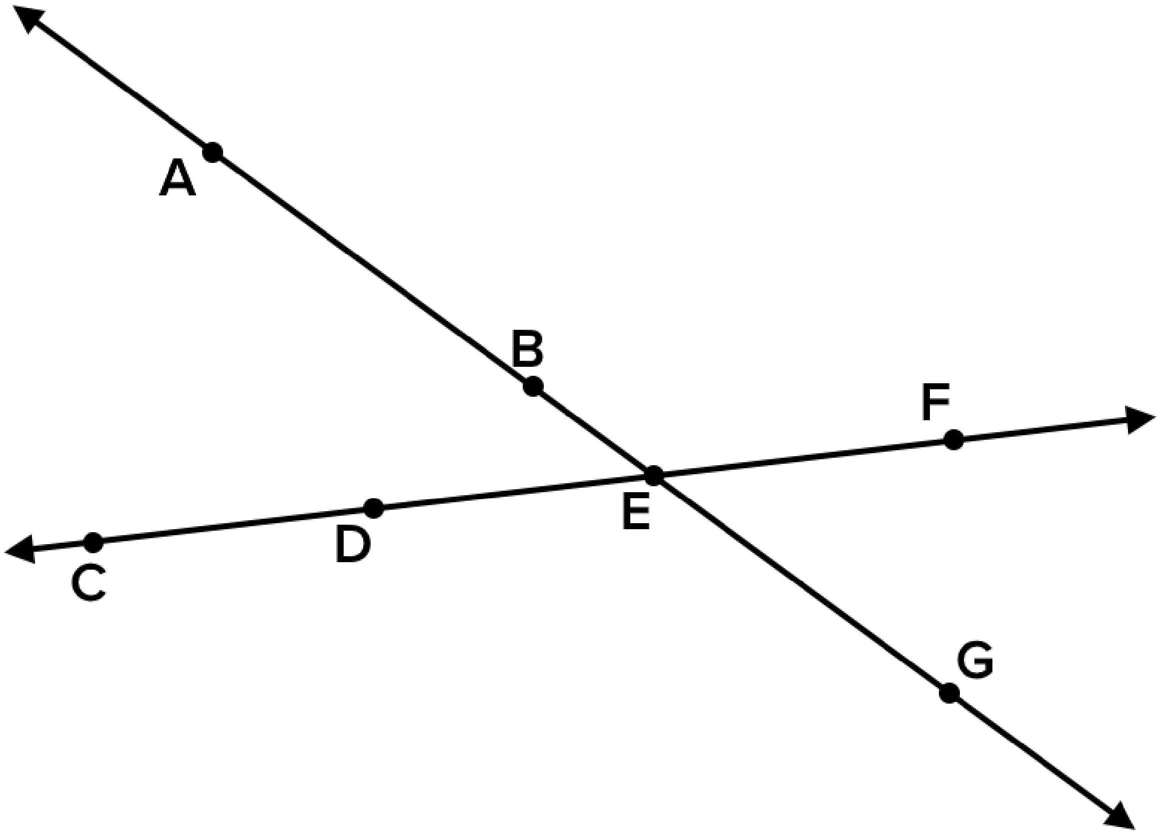


b

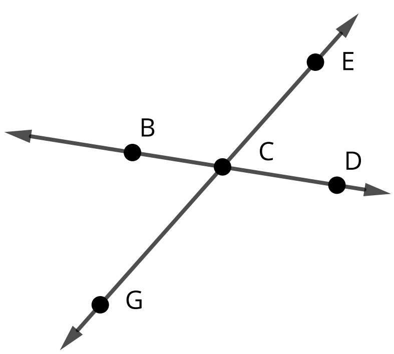


3 What is another way to define the line $\overleftrightarrow{CD}$?

4 What is another way to describe the same s[illegible]points as $\overline{FD}$?



5 Determine whether the statement is true or false based on the diagram.



a Point C is on $\overleftrightarrow{GE}$.

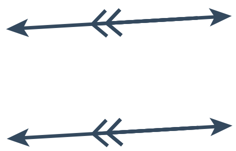
b Point C is on $\overleftrightarrow{BD}$.

c The intersection of $\overleftrightarrow{GE}$ and $\overleftrightarrow{BD}$ is point C.

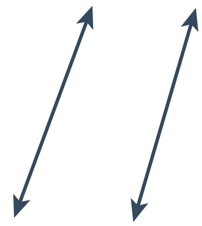
d $\overleftrightarrow{GE}$ and $\overleftrightarrow{BD}$ must be perpendicular because they intersect.

6 In each diagram below, state whether the lines are parallel or not parallel.

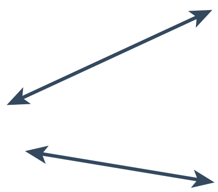
a



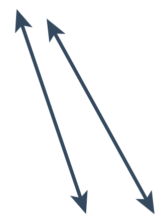
b



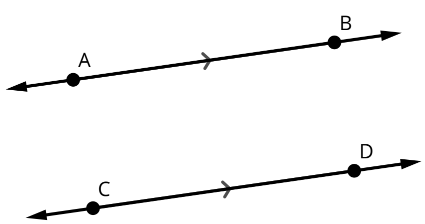
c



d



7 Consider the diagram below.



State whether each of the following statements is true, false, or if there is not enough information to decide.

a $\overline{AB} \parallel \overline{CD}$

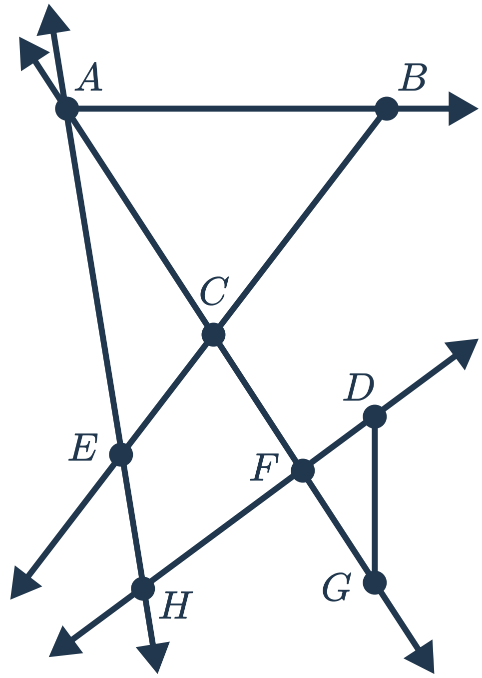
b $\overline{AB} \cong \overline{CD}$

c $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$

d $\overrightarrow{AB} \parallel \overrightarrow{CD}$

Additional questions

8 Using proper notation, identify the following features of the diagram shown:



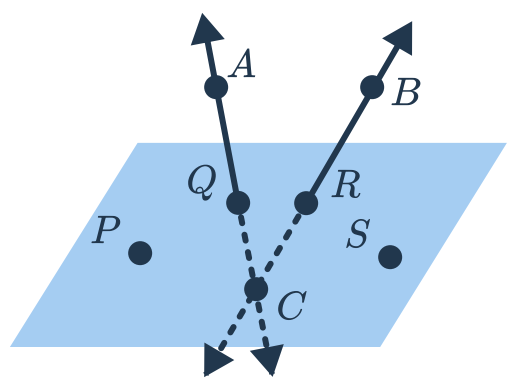
a A set of three collinear points

b A line passing through point F

c A ray starting from point A

d Another name for the ray $\overrightarrow{BE}$

9 Using proper notation, identify the following features of the diagram shown:



a A line that is contained in plane PQR

b A line that intersects, but is not contained in, plane PQR

c Four coplanar points

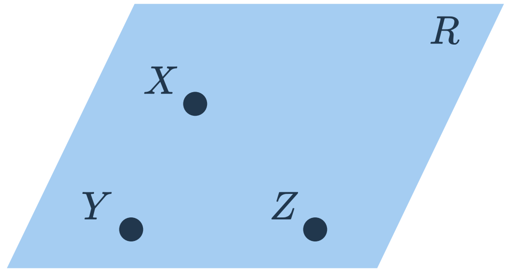
d Three collinear points

e An angle that is contained in the plane

f An angle that is not contained in the plane

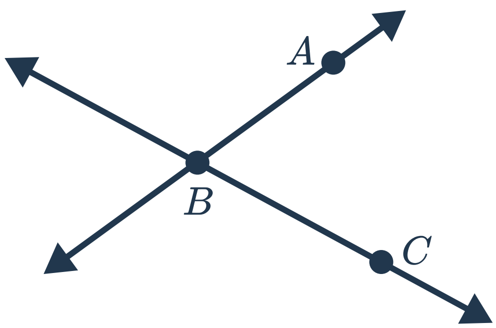
g A pair of opposite rays

10 Name the geometric figure shown below in [redacted]ifferent ways:

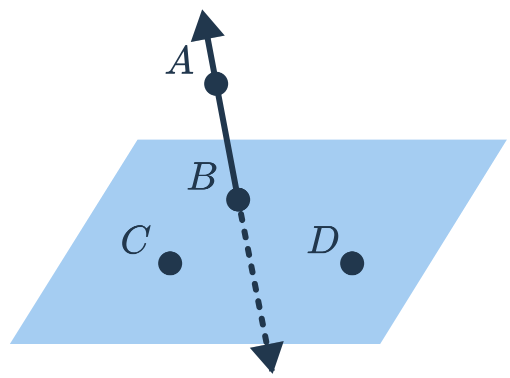


11 State the intersection of the two geometric figures shown in each of the following diagrams:

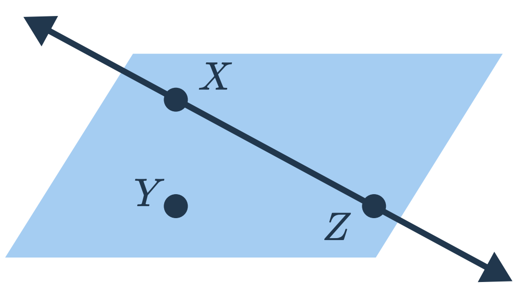
a Lines $\overleftrightarrow{AB}$ and $\overleftrightarrow{BC}$



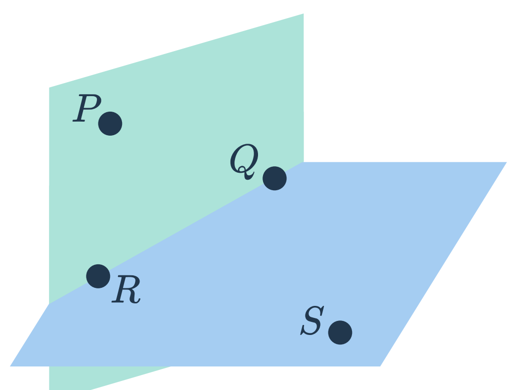
b Line $\overleftrightarrow{AB}$ and Plane BCD



c Line $\overleftrightarrow{XZ}$ and Plane XYZ



d Planes PQR and QRS



12 Answer the following questions:

a Draw a ray that has point A as an endpoint.

b Write a name for the ray you have drawn.

c How many points did you need to use to name the ray?

d When naming the ray, does the order of the points matter? Explain your answer.

13 Lines $\overleftrightarrow{PQ}$ and $\overleftrightarrow{RS}$ meet at point M.

a Draw a diagram that shows this.

b Identify a pair of opposite rays in the diagram.

14 Planes DEF and EFG have common points of E and F.

a Draw a diagram that shows this.

b Name the full intersection of the two planes.

15 Draw a diagram that shows $\overline{AB}$ contained in plane P, with $\overleftrightarrow{CB}$ intersecting (but not conatined in) the plane.

16 Determine whether each of the following statements are true or false. If true, draw an example. If false, draw a counter example.

a Through any two points, there is exactly one line.

b Two distinct pairs of points always form two different lines.

c If two points are both contained in a plane, then the line that they form is also entirely contained in that plane.

d For any three collinear points, there is exactly one plane that contains them.

e If three points are collinear, they cannot be contained in a single plane.

f For any two distinct lines, there is a plane which contains them both.

17 For each of the following geometric figures, determine a real life example which is a model for the figure. Are there any limitations or restrictions on the real life example?

a A ray b A line c A plane d Skew lines